# PROJECT CASE STUDY

UAE RETAIL & FMCG BUSINESS INTELLIGENCE SYSTEM

SQL SERVER • SQL • POWER BI • DAX • AED • 2025

From separate retail datasets to a consolidated, interactive management report.

**Project type:** End-to-end data analytics and business intelligence portfolio project

**Business scenario:** A fictional UAE multi-store retailer with a broad FMCG and general merchandise assortment.

**Data transparency: This is a synthetic/generated portfolio dataset, not audited information from a real UAE retailer.**

01 · EXECUTIVE SUMMARY

# The case in one page

Retailers generate information across sales, stores, products and operational processes. Reviewing these records separately makes it harder to understand overall performance and identify where management attention is needed.

## Business challenge

Management needs a consistent view of sales and gross profitability across stores, categories, channels and months, alongside supporting views of inventory, wastage and customers.

## Solution

A synthetic multi-store retail dataset was imported into SQL Server, checked and analysed with SQL, then modelled and presented through a multi-page Power BI dashboard with DAX measures, slicers and sidebar navigation.

## Outcome

The report consolidates headline KPIs and provides supporting pages for store, product, inventory, wastage and customer analysis. It helps users identify patterns and decide where to investigate further.

| Headline KPI | Current Overview result |
|---|---|
| Revenue | Approximately AED 14.97 million |
| Gross Profit | Approximately AED 4.26 million |
| Gross Profit % | 28.47% |
| Transactions | Approximately 100,000 |
| Average Basket Value | AED 149.70 |

**Core value: The project demonstrates the full analytics workflow—business understanding, data preparation, validation, SQL analysis, KPI design, dashboard development and business interpretation.**

02 · BUSINESS CONTEXT

# Problem statement and objectives

| Problem area | Why it matters | Project response |
|---|---|---|
| Fragmented reporting | Separate datasets make consistent comparisons difficult. | Organized analytical data and a consolidated report. |
| Store visibility | Managers need to compare location performance. | Store-level performance views. |
| Product mix | Revenue alone does not show profitability or product contribution. | Category and product analysis. |
| Inventory risk | Low, excess or slow-moving stock can affect availability and working capital. | Inventory-focused analysis using available fields. |
| Wastage | Recorded product losses reduce value retained by the business. | Wastage breakdowns by available dimensions. |
| Customer understanding | Sales totals do not fully explain customer or basket patterns. | Customer and basket analysis where the data supports it. |

## Project objectives

- Create a realistic UAE retail reporting scenario using AED and multiple store locations.
- Store and inspect the datasets in SQL Server.
- Validate data quality and key calculations before reporting.
- Use SQL to answer business questions and cross-check headline figures.
- Build reusable Power BI measures and a navigable multi-page report.
- Translate patterns into practical questions for management.

03 · DATA AND TECHNOLOGY

# Dataset, tools and structure

The portfolio dataset represents a multi-store retail operation over a 2025 reporting period.

| Dataset characteristic | Project detail |
|---|---|
| Stores | 9 |
| Products | 5,001 |
| Reporting period | 2025 |
| Currency | AED |
| Transactions | Approximately 100,000 as displayed in the Executive Overview |
| Data source | Synthetic/generated CSV files |

## Main data areas

| Data area | Example tables | Analytical role |
|---|---|---|
| Master data | Stores, Products, Categories | Describe and group stores and products. |
| Sales | Sales, Sales_Items | Bill-level transactions and product lines. |
| Customer | Customers and sale customer IDs | Support customer and basket analysis where populated. |
| Operations | Purchases, Inventory_Snapshots, Returns, Wastage | Support inventory and loss analysis. |
| Other reference data | Targets, Employees | Provide target or operational context where used. |

## Technology stack

- **SQL Server / SSMS:** database storage, table inspection, SQL queries and validation.
- **SQL:** joins, aggregations, data checks and business-question analysis.
- **Power Query:** preparing and loading data into Power BI.
- **Power BI:** relationships, report pages, slicers and visual exploration.
- **DAX:** reusable KPI calculations that respond to report filters.

**Important data-grain concept: Sales is at bill/transaction level; Sales_Items is at product-line level. Count distinct sale IDs for transactions—do not count item rows as bills.**

04 · METHODOLOGY

# End-to-end project process

The workflow moves from business requirements to validated, decision-support reporting.

| Stage | What was done | Why it matters |
|---|---|---|
| 1. Business understanding | Defined questions about sales, profit, stores, categories, inventory, wastage and customers. | Ensures the report answers real business questions. |
| 2. Dataset preparation | Used generated CSV datasets for retail master and transaction data. | Creates a connected multi-area retail scenario. |
| 3. SQL Server import | Imported CSV files into database tables and inspected their structure. | Centralizes data for querying and validation. |
| 4. Data-quality review | Reviewed row counts, data types, keys, missing values, dates and numeric fields. | Reduces the risk of incorrect calculations. |
| 5. SQL analysis | Queried and summarized sales and operational data. | Answers business questions and provides validation totals. |
| 6. Power BI model | Loaded relevant tables and created relationships. | Supports consistent filtering and aggregation. |
| 7. DAX measures | Created measures for revenue, profit, margin, transactions and ABV. | Keeps KPI calculations reusable and consistent. |
| 8. Dashboard design | Built the Overview and supporting analysis pages with sidebar navigation and slicers. | Makes information easier for managers to explore. |
| 9. Validation & interpretation | Compared headline Power BI values with SQL outputs and interpreted visible patterns. | Improves confidence and communicates next steps. |

**Interview explanation: “I checked the headline Power BI figures against SQL results before relying on the visuals. This helped me confirm that the report used consistent calculations.”**

05 · DASHBOARD DESIGN

# Six report pages and navigation

The dashboard uses a consistent teal theme, a left-side sidebar and slicers such as Month, City, Stores and Channel.

| Page | Purpose | Business question |
|---|---|---|
| Overview | Revenue, gross profit, margin, transactions and ABV, plus category, channel, store and monthly visuals. | How is the business performing overall? |
| Stores | Store comparisons and performance measures. | Which locations contribute most or need review? |
| Products | Category and product contribution. | Which products and categories drive revenue? |
| Inventory | Stock indicators and stock-related analysis where supported by data. | Where may low, excess or slow-moving stock exist? |
| Wastage | Recorded losses and their patterns. | Where are losses concentrated? |
| Customers | Customer and basket analysis where supported by the fields. | What can the data tell us about purchasing patterns? |

## User interaction

- Sidebar buttons navigate between report pages.
- Slicers narrow the report by month, city, store and channel.
- Charts compare performance by store, category, channel and month.
- A consistent KPI layout helps users interpret the same measures across pages.

**Final scope: Suppliers and Promotions pages were removed. Do not describe them as completed pages in the final dashboard; they may be mentioned as possible future extensions.**

06 · FINDINGS

# What the current Overview displays

The following findings are supported by the current Executive Overview screenshot and project details. Figures are rounded.

| Visible finding | Evidence shown | Business interpretation |
|---|---|---|
| Overall performance | Revenue AED 14.97M; gross profit AED 4.26M; GP% 28.47%; around 100K transactions; ABV AED 149.70. | Provides the headline scale, gross profitability and average transaction value. |
| Leading category | Beverages is shown at approximately AED 1.84M revenue. | A major revenue contributor; check its margin and stock position before making range decisions. |
| Highest visible store bars | Marina Grand AED 2.42M; Khalidiya AED 2.28M; Al Barsha AED 2.20M. | These lead the visible revenue chart, but revenue alone does not establish the best overall store. |
| Channel mix | Store channel about AED 13.56M or 90.6% of revenue. | The displayed business is strongly store-led; the chart does not explain why other channels are smaller. |
| Monthly view | January–December monthly trend is displayed. | Managers can identify periods for investigation alongside seasonality, stock and other events. |

**Evidence boundary: The screenshot shows patterns, not their causes. It does not prove that the dashboard increased sales, reduced wastage or improved customer satisfaction.**

07 · BUSINESS USE

# How management can act on the report

| Signal / question | Possible action | Follow-up measure |
|---|---|---|
| High revenue but lower margin at a store | Review product mix, discounts, purchase costs and pricing. | Store gross profit, GP% and sales mix. |
| Important category contribution | Check margin, availability and replenishment priorities. | Category revenue, GP%, stock cover and out-of-stock rate. |
| Sales decline in a store or month | Investigate transactions, ABV, channel mix and availability. | Comparable-period revenue, transactions and ABV. |
| Inventory exceptions | Review reorder quantities, replenishment and slow-moving products. | Stock cover, ageing, inventory value and stockouts. |
| Wastage concentration | Investigate expiry, handling, ordering or storage where relevant. | Wastage value and rate by reason, SKU and store. |
| Customer/basket differences | Review assortment and customer engagement opportunities. | Customer and basket measures supported by actual data. |

## Expected value

- Less manual consolidation and quicker access to key performance information.
- More consistent discussions using shared KPI definitions.
- Better-targeted investigations into stores, products, stock and wastage.
- A way to track whether actions improve results over time.

**Business value statement: The report supports visibility and better-informed decisions. Actual financial improvements or time savings must be measured after the retailer acts.**

08 · LIMITATIONS AND NEXT STEPS

# What the project does—and does not—prove

| Area | Current limitation | Possible future improvement |
|---|---|---|
| Dataset | Synthetic/generated, not real retailer records. | Validate on approved real business data. |
| Causality | Charts show patterns but do not prove causes. | Add promotion calendars, pricing, stock events and operational context. |
| Customer experience | No direct satisfaction or service outcome is established. | Add survey, complaint, loyalty or service data if available. |
| Forecasting | A historical trend is not a validated forecast. | Build and back-test a demand forecasting model. |
| Suppliers / promotions | These pages were removed from the final report. | Add later only with validated source data and a clear business need. |

## Closing summary

**This project demonstrates how a data analyst can translate a retail business problem into structured data, validate it with SQL, build reusable Power BI KPIs and communicate findings through an interactive report. The strongest portfolio value is the end-to-end process and the ability to explain both what the numbers show and what they do not prove.**

## Interview-ready description

“I built a UAE Retail & FMCG BI portfolio project using synthetic data for 9 stores and 5,001 products. I used SQL Server for storage, data checks and analysis, then Power BI and DAX to create a six-page interactive report. The Overview displays approximately AED 14.97M revenue, AED 4.26M gross profit, 28.47% gross profit margin, around 100,000 transactions and AED 149.70 average basket value. The report helps management compare stores, categories and channels and investigate inventory and wastage patterns. I clearly disclose that the data is synthetic and that the dashboard supports decisions rather than proving their impact.”